

Lv 1: Recognition (Concept)

Visual Concept Perception

Textual Context: There are four plates on the counter, all laden with cream puffs, creating a tempting display of pastries.
Question: How many plates are there on the counter?
Answer: 4

Visual Context

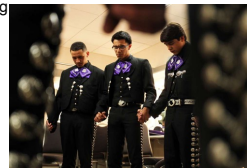


Lv 2: Understanding (Relation)

Visual Scene Comprehension

Textual Context: The three men are standing closely together, each holding the hand of the person next to them.
Question: What interaction is taking place between the three men in the image?
Answer: They are holding hands.

Visual Context

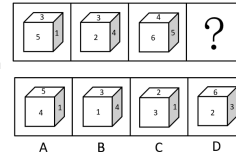


Lv 3: Reasoning (Thinking)

Deep Visual Reasoning

Textual Context: All dice in the sequence are the same die, just viewed from different angles.
Question: Which of the following options is most likely to be this cube in the image?
Answer: C.

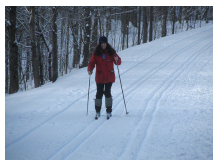
Visual Context



Basic Cross-modal Association

Textual Context: A woman is seen cross-country skiing up a snow-covered hill, surrounded by a forest of bare trees on a winter day. She is wearing a red jacket, dark pants, and a blue knit hat. She is holding ski poles, but she doesn't have skis on her feet.
Question: Is the image context aligned with the text?
Answer: The woman is wearing skis on her feet, but the text states she doesn't have skis on her feet.

Visual Context



Integrative Cross-modal Understanding

Textual Context: The objects on the desk, in order from left to right, are located at Position 1, Position 2, Position 3, and Position 4.
Round 1: Swap the object at Position 1 with the object at Position 3.
Round 2: Swap the apple with the object that was initially at Position 2.
Question: After all swaps, what is the final position of the Rubik's Cube?
Answer: Position 1

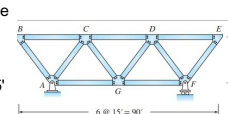
Visual Context



Deep Cross-modal Reasoning

Textual Context: The truss is to be designed for a uniform live load of 0.32 kip/ft that can be placed anywhere on the span, in addition to a concentrated live load of 24 kips that can be positioned where it will produce the largest force in bar CG. The truss has a span of 6 @ 15' = 90' and a height of 20'.
Question: Determine the maximum value of live load force created in bar CG.
Answer: 9.75kips compression

Visual Context



Linguistic Concept Processing

Textual Context: Sentence 1: Your wish will come true. Sentence 2: The water came to a boil.
Question: Does the word 'come' (verb) have the same meaning in both sentences?
Answer: Yes

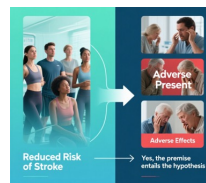
Visual Context



Contextual Comprehension

Textual Context: Premise: After two years, the aspirin subjects experienced a reduced risk of fatal and nonfatal stroke. Hypothesis: People experienced adverse effects while taking aspirin.
Question: Does the premise entail the hypothesis?
Answer: Yes, the premise entails the hypothesis

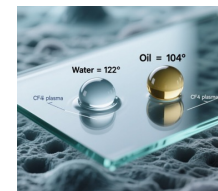
Visual Context



Deep Textual Reasoning

Textual Context: The water and oil contact angles on a smooth clean glass surface are 65° and 40° respectively. The surface of this same piece of glass is then modified by treatment with a CF4 plasma.
Question: What would be the best estimate of the water and oil contact angles on the treated surface?
Answer: Water = 122°, Oil = 104°

Visual Context



Vision-Centric

Synergy

Text-Centric